2020



Physics

دعاء بداية المذاكره :

اللهم إني اسألك فهم النبين ، وحفظ الملائكة المقربين،
وأن تجعل لساني عامراً بذكرك ، وقلب بخشيتك ،
وبدني بطاعتك فأنت حسبي ونعم الوكيل .

WRITEN BY ISLAM SHAKER

STEM-OBOUR

LO - 1

All Laws

Gravitational Force:

$$F_g = G\,\frac{m_1 m_2}{r^2}$$

Gravitational Constant

$$G = 6.67 \times 10^{-11}\ \ N.\frac{m^2}{kg^2}\ \ or\ \ \frac{m^2}{s^2}.kg^2\ \ or\ \ J.\frac{m}{kg^2}$$

Free-Fall Acceleration:

$$g = \frac{GM}{r^2} = \frac{GM}{(R_E+h)^2}$$

Gravitational Field:

$$\vec{g} = \frac{\overrightarrow{f_g}}{m}$$

Gravitational Potential Energy:

$$U = -\frac{Gm_1 m_2}{r}$$

Escape Speed:

$$V_{esc} = \sqrt{\frac{2GM}{r}} = \sqrt{2gr}$$

Orbit Velocity:

$$v_o = \sqrt{\frac{GM}{r_o}}$$

Centripetal acceleration: $a_c = \frac{v^2}{r}$ / **Centripetal Force:** $F_c = \frac{mv^2}{r}$

Mass of the Earth: $5.97 \times 10^{24} kg$ / **Radius of the Earth:** $6.37 \times 10^6\, m$

Notes

***Newton's Law of Universal Gravitation*:** every particle in the Universe attracts every other particle with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

By Newton's third law $\overrightarrow{F_{12}} = -\overrightarrow{F_{21}}$

When a particle is placed at a point where the gravitational field exists, the particle experiences a gravitational force. In other words, we imagine that the field exerts a force on the particle rather than consider a direct interaction between two particles.

The Gravitational Field: It equals the gravitational force $\boldsymbol{F_g}$ experienced by a test particle placed at that point divided by the mass $\boldsymbol{m}$ of the test particle.

The Escape Speed: It is the minimum speed the object must have at the Earth's surface to approach an infinite separation distance from the Earth.

Traveling at this minimum speed, the object continues to move farther and farther away from the Earth as its speed asymptotically approaches zero.

LO – 2

All Laws

Coulomb's Law:

$$F_e = k_e \frac{|q_1||q_2|}{r^2}$$

Coulomb Constant:

$$k_e = \frac{1}{4\pi\epsilon_o} = 9 \times 10^9 \ N.m^2/C^2$$

Permittivity of Free Space:

$$\epsilon_o = 8.85 \times 10^{-12} \ C^2/N.m^2$$

Electric Force:

$$\overrightarrow{F_e} = q\,\vec{E}$$

Electric Field:

$$\vec{E} = \frac{\overrightarrow{F_e}}{q} = \frac{k_e|q|}{r^2} = \frac{\Delta v}{d} \qquad (N/C \ or \ v/m)$$

The Work done by The Electric Field on The Charge:

$$w = \overrightarrow{F_e}.\overrightarrow{ds} = |q|\vec{E}.\overrightarrow{ds} = -\Delta PE = -q\Delta v$$

Electric Potential Energy:

$$\Delta PE\ (\Delta U) = -w = -q\int \vec{E}.\vec{ds} = q\Delta v = k_e\frac{q_1q_2}{r}$$

Potential Difference:

$$\Delta v = \frac{\Delta PE}{q} = -\int \vec{E}.\overrightarrow{ds} = k_e\frac{q}{r} \quad (V = J/C)$$

Electric Potential Energy of a Pair Charges:

$$U = k_e\frac{q_1q_1}{r}$$

Notes

Electrons are negatively charged, and **Protons** are positively charged.

Charges of the same sign **repel** one another.

Charges with opposite signs **attract** one another.

Electric Charge is always **conserved** in an isolated system.

The Electrified State: is due to a transfer of charge from one object to the other.

One object gains some amount of **negative charge** while the other gains an equal amount of **positive charge.**

Electrical Conductors: Materials in which some of the electrons are free electrons that are not bound to atoms and can move relatively freely through the material.

Electrical Insulators: Materials in which all electrons are bound to atoms and cannot move freely through the material.

Semiconductors: Materials their electrical properties are somewhere between those of insulators and those of conductors.

The electrical properties of semiconductors can be changed over many orders of magnitude by the addition of controlled amounts of certain atoms to the materials.

There are Three Types of Charging:

Charging by Conduction; in which occurs contact among objects.

Charging by Induction; in which occurs no contact among objects (Metals).

Charging by Polarization; in which occurs no contact among objects (Nonmetals).

Coulomb's law: States that the electric force which exerted by a source charge (q_1) on a test charge (q_2) is directly proportional to the product of two charges and inversely proportional to the square of the distance between them.

If q_1 and q_2 have the same sign the product $(q_1\ q_2)$ is positive and the electric force on one particle is directed **away** from the other particle. **If q_1 and q_2** have the opposite sign the product $(q_1\ q_2)$ is negative and the electric force on one particle is directed **toward** the other particle.
Electric Field: It defined as the electric force F_e that acts on a small positive test charge placed at that point divided by the magnitude $q_o$of the test charge.
If q is positive, the force is in the same direction as the field. **If q is negative,** the force and the field are in opposite directions.
For a positive source charge, the electric field at Point charge radially outward. **For a negative source charge,** the electric field at point charge radially inward.
The Density of field lines through surface is inversely proportional to the area of the surface.
The Number of lines per unit area on the sphere is $(N/4\pi r^2)$
The lines must begin on **a positive** charge and terminate on **a negative** charge. In the case of an excess of one type of charge, some lines will begin or end infinitely far away.
The Electric Dipole: It is electric field lines for two points charges of equal magnitude but opposite signs.
The Number of lines that begin at the **positive** charge must equal the number that terminate at the **negative** charge.
When a charge q is placed in an electric field $\vec{E}$ created by source charge distribution, the particle in a field model tells us that there is an electric force $q\vec{E}$ acting on it.
The Electric Field will be doing work on the charge; This work is internal to the system.

Internal Work done in a system is equal to the negative of the change in the potential energy of the system.
The Electric Potential: It is dividing the potential energy by the charge and depends only on the source charge distribution.
The Potential Difference: It is the change in electric potential energy of the system when a charge q is moved between the points divided by the charge.
The electric field is a measure of the rate of change of the electric potential with respect to position.
All Points on the surface of a charged conductor in electrostatic equilibrium are at the same potential. **The Electric Potential** is a constant everywhere on the surface of a charged conductor in equilibrium. **The Electric Potential** is constant everywhere inside a conductor and equal to that same value at the surface.
Electric field lines always point in the direction of decreasing electric potential.
In a System consisting of a positive charge and an electric field, the electric potential energy of the system decreases when the charge moves in the direction of the field. **In a system** consisting of a negative charge and an electric field gains electric potential energy when the charge moves in the direction of the field. **For the Negative Charge** to move in the direction of the field, an external agent must apply a force and do positive work on the charge.
The Equipotential Surface: It is any surface consisting of a continuous distribution of points having the same electric potential.
Superposition Principle: It is the total electric potential at some point P due to several point charges is the sum of the electric potentials due to the individual charges.

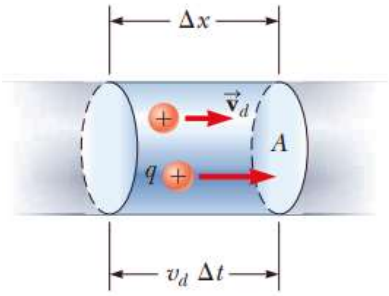
LO - 3

All Laws

Average Current:

$$I_{avg} = \frac{\Delta Q}{\Delta t} = nqv_d A \quad \left(A = \frac{C}{s} \right)$$

$$\Delta Q = (n\,A\,\Delta x)q = (n\,A\,v_d\,\Delta t)q$$



$\Delta Q =$ The amount of charge that passes through the surface.

$\Delta t =$ a time interval.

$n =$ The number of charge carriers per unit volume.

$A =$ The cross-section area of the wire.

$v_d =$ A velocity of the charge that parallel to the axis of the wire.

$\Delta x =$ the displacement or the length of the wire.

The Current Density:

$$J = \frac{I}{A} = nqv_d = \sigma E = \sigma \frac{\Delta v}{l} \quad (A/m^2)$$

Ohm's Law or Potential Difference:

$$\Delta v = El = \frac{l}{\sigma} J = \left(\frac{l}{\sigma A} \right) I = IR$$

Resistance:

$$R = \frac{\Delta v}{I} = \frac{l}{\sigma A} = \rho \frac{l}{A} \qquad (\Omega = v/A)$$

$\Delta v =$ Potential Difference

$I =$ Electric Current

$l =$ The length of the wire

$A =$ cross-sectional area of the wire

$\sigma =$ Conductivity of the material

$\rho =$ Resistivity of the material

Variation of R with temperature:

$$R = R_o[1 + \alpha(T - T_o)]$$

$R =$ The Resistance

$\alpha = \frac{1}{R_o} . \frac{\Delta R}{\Delta T}$ (temperature coefficient of resistivity)

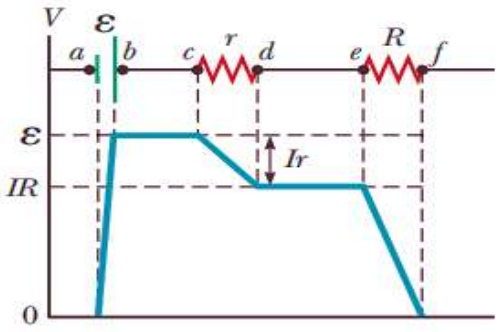
$T =$ Temperature

Electrical Power:

$$P = I\,\Delta v = I^2 R = \frac{\Delta v^2}{R} \quad (W = A.V)$$

Terminal voltage:

$$\Delta V = \varepsilon - Ir$$



Electromotive Force:

$$\varepsilon = IR + Ir$$

The Current:

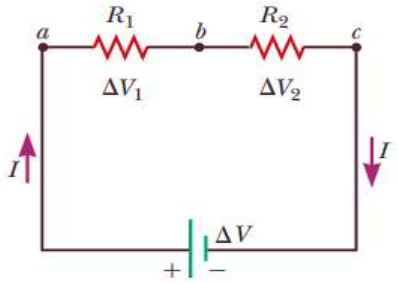
$$I = \frac{\varepsilon}{R+r}$$

Resistors in Series Combination:

$$I = I_1 = I_2$$

$$\Delta v = \Delta v_1 + \Delta v_2$$

$$R_{eq} = R_1 + R_2$$



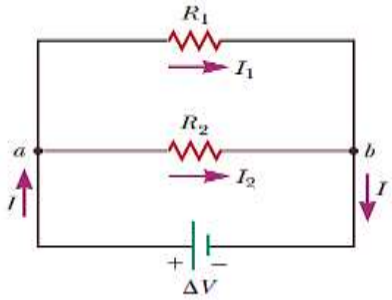
The Equivalent Resistant is always Greater than any Individual Resistance.

Resistors in Parallel Combination:

$$\Delta v = \Delta v_1 = \Delta v_2$$

$$I = I_1 + I_2$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$



The Equivalent Resistance is always less than The Smallest Resistance.

Notes

The amount of flow depends on both the **material** through which the charges are passing and the **potential difference** across the material.

To define current quantitatively, suppose charges are moving perpendicular to a surface area.

The Current: It is the rate at which charge flows through the surface.

If ΔQ is the amount of charge that passes through the surface in a time interval Δt, the average current is equal to the charge that passes through surface per unit time

The Current has the same direction as the flow of positive charge.

In Electrical Conductors, the current results from the motion of negatively charged electrons.

In an Ordinary Conductor, the direction of the current is opposite the direction of flow of electrons.

In some cases—such as those involving gases and electrolytes, the current is the result of the flow of both positive and negative charges.

If the Ends of a Conducting Wire are connected to form a loop, all points on the loop are at the same electric potential; hence, the electric field is zero, there is no net transport of charge through the wire; therefore, there is no current.

If the Ends of a Conducting Wire are connected to a battery, all points on the loop are not at the same potential; hence, potential difference between the ends of the loop, an electric field. The electric field exerts forces on the electrons in the wire, causing them to move in the wire and therefore creating a current.

The Drift Speed: It is the speed of the charge carriers v_d is an average speed.

When a potential difference is applied across the conductor an electric field is set up; This field exerts an electric force on the electrons, producing a current.

The energy transferred from the electrons to the metal atoms during collisions causes an increase in the atom's vibrational energy and a corresponding increase in the conductor's temperature.

The current density: J in the conductor is defined as the current per unit area.

A current density is established in a conductor whenever a potential difference is maintained across the conductor. In some materials, the current density is proportional to the electric field.

For many materials, the ratio of the current density to the electric field is a constant $\boldsymbol{\sigma}$ that is independent of the electric field producing the current.

Ohmic Materials: which obey Ohm's law and, hence demonstrate this simple relationship between electric field $\boldsymbol{E}$ and current density $\boldsymbol{J}$. $(\boldsymbol{J = \sigma E})$

Ohmic materials have a linear current–potential difference relationship.

Nonohmic Materials: which don't obey Ohm's law, hence don't demonstrate this simple relationship between electric field $\boldsymbol{E}$ and current density $\boldsymbol{J}$. $(\boldsymbol{J = \sigma E})$

Nonohmic materials have a nonlinear current–potential difference relationship.





The Resistance: As the ratio of the potential difference across a conductor to the current in the conductor.

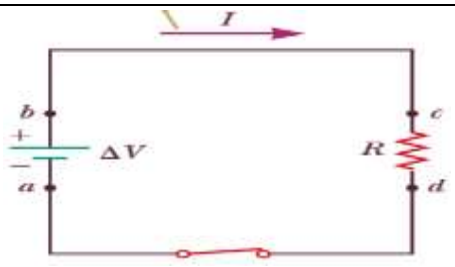
Resistor: which use to control the current in the various parts of the circuit.

An ideal conductor would have zero resistivity.

An ideal insulator would have infinite resistivity.

The Resistance of the wire is directly proportional to the **length** of the wire and inversely proportional to the cross-sectional **area** of this wire.

As the charge moves from **a** to **b** through the battery, the electric potential energy of the system increases by an amount $\boldsymbol{Q\ \Delta V}$ while the chemical potential energy in the battery decreases by the same amount.

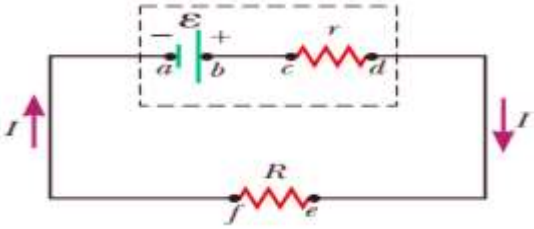


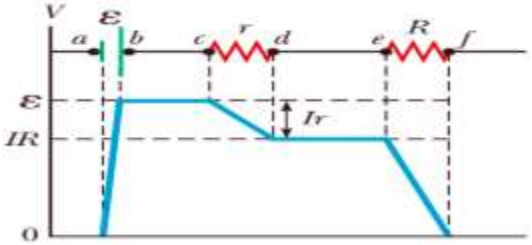
As the charge moves from **c** to **d** through the resistor, however, the electric potential energy of the system decreases due to collisions of electrons with atoms in the resistor.

The Electromotive Force (emf ε) of a battery: It is the maximum possible voltage the battery can provide between its terminals.

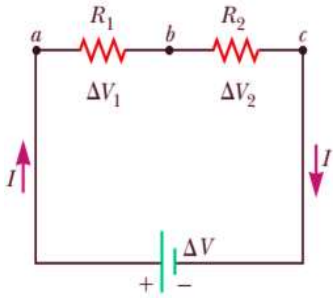
When an electric potential difference exists between two points, the source moves charges from **the lower** potential to **the higher**.

For an idealized battery with zero internal resistance, the potential difference across the battery equals its **emf**.





Resistors in a Series Combination:

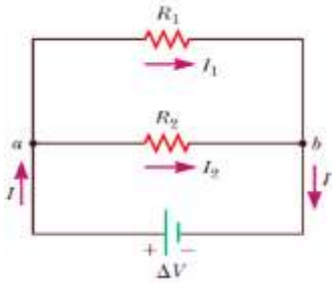


The Amount of Charge is the same in all resistors.

The Potential Difference divides between the resistors.

The Equivalent Resistance is the numerical sum of the individual resistances and is always greater than any individual resistance.

Resistors in a parallel Combination:



The Amount of Charge divides between the resistors.

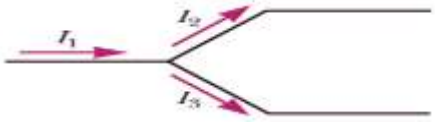
The Potential Difference is the same in all resistors.

The Equivalent Resistance is equal to the sum of the inverses of the individual resistances and is always less than the smallest resistance.

LO - 4

Junction Rule: At any junction, the sum of the currents must equal zero.

$$\left(\sum I_{junction} = 0\right)$$



Loop Rule: At any loop, the sum of the potential differences must be zero.

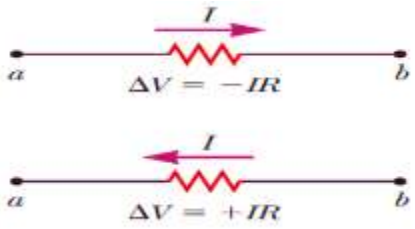
$$\left(\sum \Delta v_{Loop} = 0\right)$$

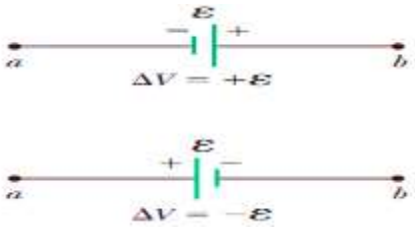
If a resistor is traversed in the direction of the current, the potential difference ΔV across the resistor is $-IR$.

If a resistor is traversed in the direction opposite the current, the potential difference ΔV across the resistor is $+IR$.

If a Source of emf is traversed in the direction of the emf (from negative to positive), the potential difference ΔV is $+\varepsilon$.

If a Source of emf is traversed in the direction opposite the emf (from positive to negative), the potential difference ΔV is $+\varepsilon$.





LO - 5

All Laws

The Capacitance:

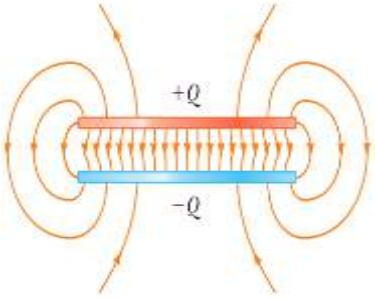
$$C = \frac{Q}{\Delta v} \quad (F = C/V)$$

Capacitance of an Isolated Charged Sphere:

$$C = \frac{Q}{\Delta v} = \frac{Q}{k_e\, Q/a} = \frac{a}{k_e} = 4\pi\epsilon_o a$$

The capacitance of a parallel-plate capacitor:

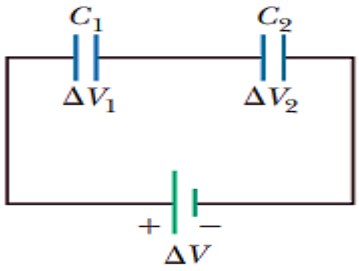
$$C = \frac{Q}{\Delta v} = \frac{\sigma A}{Ed} = \frac{\sigma A}{(\sigma/\epsilon_o)} d = \epsilon_o \frac{A}{d}$$



$E = \sigma/\epsilon_o$ / $\Delta v = Ed$ / $Q = \sigma A$

σ = The magnitude of the charge per unit area on each plate.

Parallel Combination of The Capacitors:



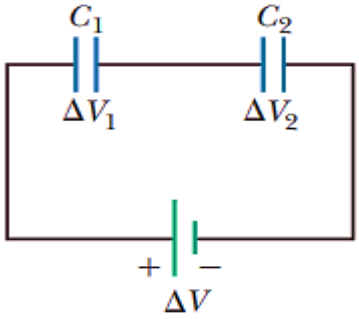
$\Delta v = \Delta v_1 = \Delta v_2$

$Q_{total} = Q_1 + Q_2$

$$C_{eq} = C_1 + C_2$$

The Equivalent Capacitance is greater than any of Individual Capacitances.

Series Combination of The Capacitors:



$Q = Q_1 = Q_2$

$\Delta v_{total} = \Delta v_1 + \Delta v_2$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$$

The Equivalent Capacitance is always less than any of Individual Capacitance.

Energy stored in a charged capacitor:

$$U_E = \frac{Q^2}{2C} = \frac{1}{2} Q\, \Delta V = \frac{1}{2} C\, (\Delta v)^2$$

Capacitance of a capacitor with a dielectric:

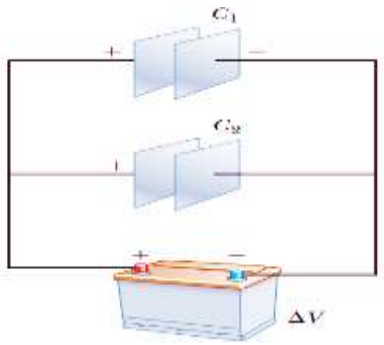
$$C = \frac{Q_o}{\Delta v_o / k} = k\, C_o = k\, \epsilon_o \frac{A}{d}$$

Notes

A capacitor: is a device used in a variety of electric circuits, it consists of two parallel metal plates separated by a distance $\boldsymbol{d}$. One plate is **negatively** charged and another is **positively** charged.

The capacitance C of a capacitor: It is the ratio of the magnitude of the charge $\boldsymbol{Q}$ on either conductor (plate) to the magnitude of the potential difference $\Delta \boldsymbol{v}$ between the conductors (plates). $\boldsymbol{SI\ unit}$: $\boldsymbol{farad}\ (\boldsymbol{F}) = (\boldsymbol{C/V})$

Capacitors in Parallel Combination:



The Left Plate, both connected to the positive terminal of the battery, so the left plates are at the same potential.

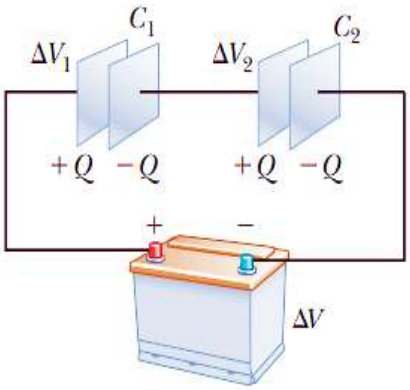
The Right Plates, both connected to the negative terminal of the battery, so the right plates are at the same potential.

Both Capacitors have the same potential difference $\Delta \boldsymbol{V}$ across them.

The Total Charge stored by the two capacitors is the sum of $\boldsymbol{Q_1}$ and $\boldsymbol{Q_2}$.

The Equivalent Capacitance is always larger than any of the individual capacitances.

Capacitors in Series Combination:



A battery is connected to the circuit, electrons with total charge $-\boldsymbol{Q}$ are transferred from the left plate of $\boldsymbol{C_1}$ to the right plate of $\boldsymbol{C_2}$ through the battery, leaving the left plate of $\boldsymbol{C_1}$ with a charge of $+Q$.

The equivalent capacitor must end up with a charge of $-\boldsymbol{Q}$ on its right plate and a charge of $+\boldsymbol{Q}$ on its left plate.

Both Capacitors have the same amount of charge on all plates.

The Total Potential Difference is the sum of Δv_1 and Δv_2.

The Equivalent Capacitance is always smaller than any individual capacitance.

Capacitors store electrical energy, and that energy is the same as the work required to move charge onto the plates. $\left(\boldsymbol{Energy\ Stored = w = \frac{1}{2}\ Q\ \Delta v^2}\right)$

A Dielectric: which is an insulating material.

When a dielectric is inserted between the plates of a capacitor, the capacitance increases.

If the dielectric completely fills the space between the plates, the capacitance is multiplied by the factor k, called the dielectric constant.

The Voltage across the plates is reduced by the factor $\boldsymbol{k}$. $\left(\Delta \boldsymbol{v} = \frac{\Delta v_o}{2}\right)$

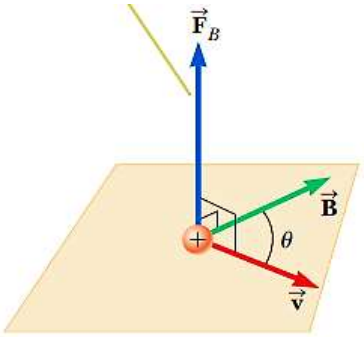
The Dielectric Strength: A maximum electric field that can be produced in the dielectric before it breaks down and begins to conduct.

LO – 6,7

All Laws

Magnetic force:

$$\vec{F} = q\vec{v}\vec{B}\sin\theta = \frac{mv^2}{r}$$

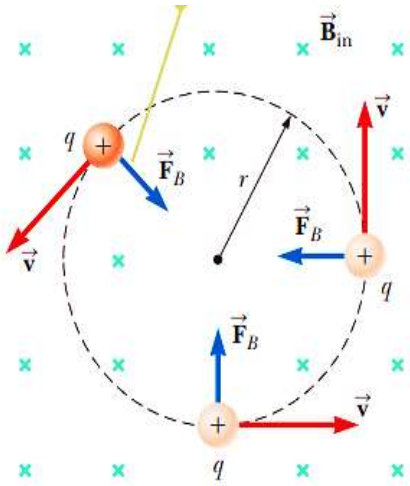


Magnetic Field:

$$\vec{B} = \frac{\vec{F}}{q\vec{v}\sin\theta} \quad \left(\frac{N}{C\ .\ m/s}\right) \textit{ or } \left(\frac{N}{A\ .\ m}\right)$$

The Radius of The Circular:

$$r = \frac{mv}{qB}$$



The Angular Speed:

$$\omega = \frac{v}{r} = \frac{qB}{m}$$

The Period of The Motion or Time Interval:

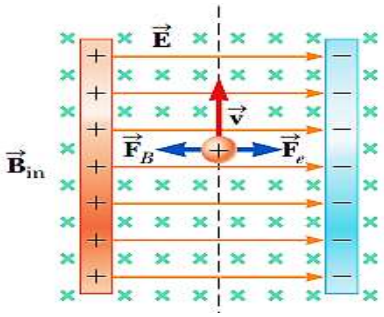
$$T = \frac{2\pi r}{v} = \frac{2\pi}{\omega} = \frac{2\pi m}{qB}$$

Lorentz force:

$$\vec{F} = q\vec{E} + q\vec{v} \times \vec{B}$$

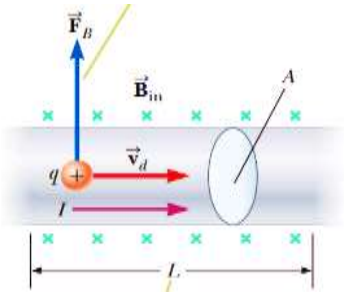
Velocity Selector:

$$v = \frac{\vec{E}}{\vec{B}}$$



Magnetic Force Acting on a Current Carrying Conductor:

$$\vec{F}_B = (q\vec{v_d} \times \vec{B})nAl = I\,L \times \vec{B}\ \sin\theta$$



Torque on a Current Loop:

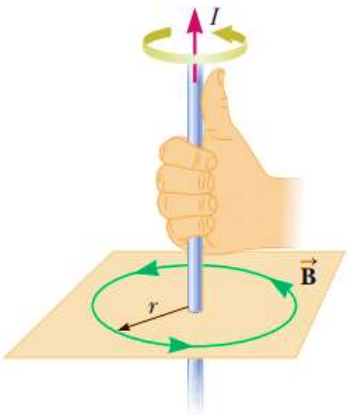
$$\tau = I\,\vec{A} \times \vec{B}\sin\theta = \mu \times \vec{B}\ \sin\theta$$

Magnetic Dipole Moment:

$$\mu = N.I\,\vec{A} \quad (A.m^2)$$

I = A Current

$\vec{A}$ = The Area of The Loop. / N = The number of the loops.



Potential energy of a magnetic moment in a magnetic field

$$U_B = -\vec{\mu} \cdot \vec{B}$$

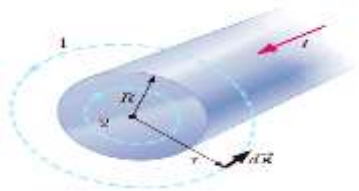
Biot–Savart law:

$$\vec{B} = \frac{\mu_o}{4\pi} \cdot \frac{I\, d\vec{S} \times \hat{r}}{r^2}$$

$\mu_o = 4\pi \times 10^{-7}\, T.m/A$ (permeability of free space)

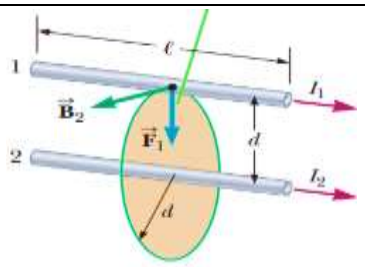
The Magnetic Field of a Long Straight Wire:

$$B = \frac{\mu_o I}{2\pi d} = \left(\frac{\mu_o I}{2\pi R^2}\right) r$$



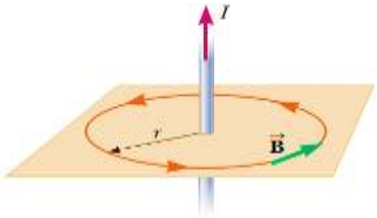
The Magnetic Force Between Two Parallel Conductors:

$$F = I\, l\, B = \frac{\mu_o I_1 I_2}{2\,\pi\, d}\, l$$



Ampere's Law:

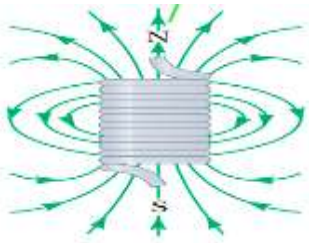
$$\vec{B}.d\vec{s} = \mu_o I$$



The Magnetic Field of a Solenoid:

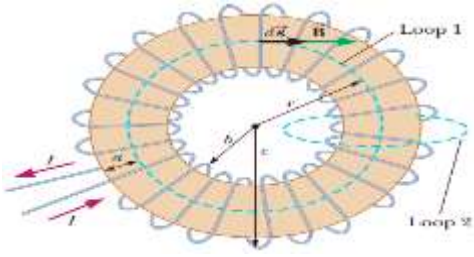
$$B = \mu_o \frac{N}{l} I = \mu_o\, n\, l$$

n = The Number of Turns per Unit Length.



The Magnetic Field of a Toroid:

$$B = \frac{\mu_o NI}{2\pi r}$$



The Magnetic Field at the Center of a Circular Loop:

$$B = N\frac{\mu_o I}{2R}$$

Moving Coil Galvanometer:

Galvanometer Sensitivity $\left(S = \frac{d\theta}{d I} = \frac{N A B}{k}\right)$

Current Density

$(Galvanometer\ Sensitivity\ for\ each\ division \times no\ of\ division)$

The current flowing $\left(I = \frac{k}{N A B}\theta\right)$

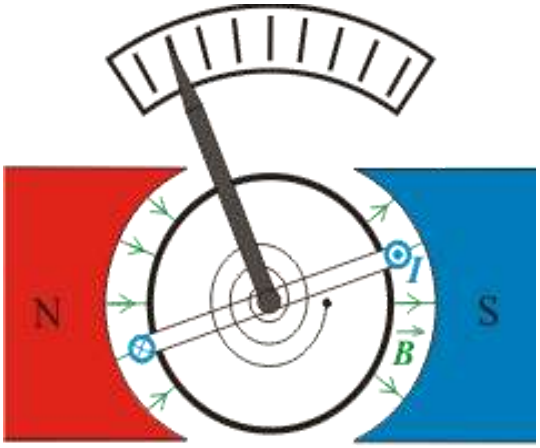
$\theta = \left(\frac{N A B}{k}\right) I$ The angle of deflection

$N =$ The number of turns

$A =$ The area of galvanometer.

$B =$ The magnetic induction.

$k =$ A couple per unit twist.



Moving Coil Ammeter:

Ammeter Resistance $\left(\grave{R} = \frac{R_g R_s}{R_g + R_s}\right)$

Ammeter Sensitivity $\left(\frac{I_g}{I} = \frac{R_s}{R_s + R_g}\right)$

Shunt Resistance $\left(R_s = \frac{I_g R_g}{I - I_g}\right)$

Moving Coil Voltmeter:

Voltmeter Resistance $\left(\grave{R} = R_g + R_m\right)$

Multiplier Resistance $\left(R_m = \frac{v - v_g}{I_g}\right)$

Notes

The Magnetic Field Lines outside the magnet point away from the **north** pole and toward the **south** pole.

When a charged particle is moving through a magnetic field, however, a magnetic force act on it.

This force has its maximum value when the charge moves in a direction perpendicular to the magnetic field lines**.**

This Force becomes zero when the particle moves along the field lines.

The strength of the magnetic force on the particle is proportional to the magnitude of the charge q, the magnitude of the velocity $\vec{v}$, the strength of the external magnetic field $\vec{B}$, and the sine of the angle θ between the direction of $\vec{v}$ and the direction of $\vec{B}$**.**

The magnetic force on a **negative** charge is directed opposite to the force on a **positive** charge moving in the same direction.

When a charged particle moves parallel to the magnetic field vector, the magnetic force on the charge is zero.

The Magnetic Force vector is perpendicular to the magnetic field.

The Magnetic Force acts on a charged particle only when the particle is in motion.

The Magnetic Force does no work when a particle is displaced in a magnetic field because the force is perpendicular to the displacement of its point.
The Magnetic Field can alter the direction of the velocity vector, but it cannot change the speed or kinetic energy of the particle.
The Radius of the Path is proportional to the linear momentum $\boldsymbol{mv}$ of the particle and inversely proportional to the charge $\boldsymbol{q}$ on the particle and the magnetic field $\vec{\boldsymbol{B}}$.
The Angular Speed of the particle and the period of the circular motion do not depend on **the speed** of the particle or on **the radius** of the orbit.
The Magnetic Force exerted on a charge $\boldsymbol{q}$ moving with a drift velocity $\boldsymbol{v_d}$ is $\left(\boldsymbol{q}\vec{\boldsymbol{v}}_{\boldsymbol{d}} \times \vec{\boldsymbol{B}}\right)$. To find the total force acting on the wire, we multiply this force by the number of charges in the segment $(\boldsymbol{n\ A\ l})$**.**
The Torque has its maximum value $(\boldsymbol{I\ A\ B}\ )$ when the field is perpendicular to the normal to the plane of the loop $(\boldsymbol{\theta = 90°})$**.**
The Magnetic Field $\overrightarrow{\boldsymbol{dB}}$ is proportional to the current $\boldsymbol{I}$ and to the length element $\overrightarrow{\boldsymbol{ds}}$. **The Magnetic Field** $\overrightarrow{\boldsymbol{dB}}$ is inversely proportional to $\boldsymbol{r^2}$ where $\boldsymbol{r}$ is the distance from $\overrightarrow{\boldsymbol{ds}}$ and $\boldsymbol{P}$**.**
Parallel Conductors Carrying currents in the same direction **attract** each other. **Parallel Conductors Carrying** currents in opposite directions **repel** each other.
If Two Long, Parallel Wires $(\boldsymbol{1\ m}\ )$ apart carry the same current and the magnetic force per unit length on each wire is $(2 \times 10^{-7} N/m)$, the current is $(\boldsymbol{1\ A})$**.**

The Magnetic Field Line has no beginning and no end. Rather, it forms a closed loop.
Ampère's law: The line integral of $\vec{B} \cdot \vec{ds}$ around any closed path equals $\mu_o I$ where I is the total steady current passing through any surface bounded by the closed path.
A solenoid: is a long wire wound in the form of a helix.
One End of the solenoid behaves like **the north pole** of a magnet and the opposite end behaves like **the south pole**. **The Length** of the solenoid increases, the interior field becomes more **uniform** and the exterior field becomes **weaker**.
Because the solenoid is ideal, $\vec{B}$ in the interior space is **uniform and parallel** to the axis and the magnetic field lines in the exterior space form **circles** around the solenoid.
A Galvanometer: An analog measuring device, denoted by **G**, that measures current flow using a needle deflection caused by a magnetic field force acting upon a current-carrying wire. **A galvanometer can function as a voltmeter** when it is connected in series with a large resistance **R**. The value of **R** is determined by the maximum voltage that will be measured. **The galvanometer can function as an ammeter** when it is placed in parallel with a small resistance **R**, often called the shunt resistance.
A Voltmeter: It measures the difference in electrical potential between two points in an electric circuit. **In order** for a voltmeter to measure a device's voltage, it must be connected in **parallel** to that device.
An Ammeter: It measures the electric current in a circuit. **In order** for an ammeter to measure a device's current, it must be connected in **series** to that device.
An Ohmmeter: It is an electrical instrument that measures electrical resistance, the opposition to an electric current.

يا رب توفيقك من حيث لا نحتسب
و افتح لنا فتحاً مبيناً